

IEEE P3335 Standard

IEEE-3335 Working Group

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1. Overview (Informative)

This document defines the architectural framework, performance expectations, and interoperability objectives for **TimeCard** devices; modular timing subsystems that provide standardized, high-precision time, phase, and frequency services to a host system.

The primary purpose of this specification is to establish a consistent structural and logical framework that accommodates various hardware implementation approaches (e.g., PCIe add-in cards, deeply embedded silicon IP, or external modules) while facilitating broad industry interoperability. This interoperability enables diverse TimeCard implementations to be seamlessly swapped, with minimal to no changes required in the host system, empowering the scalable deployment of precision timing in hyperscale computing, telecommunications, and distributed Artificial Intelligence (AI) infrastructure.

As an introductory and informative clause, the concepts presented herein provide context for the standard and do not contain normative conformance requirements.

1.1 Scope

The **IEEE P3335 TimeCard Specification** establishes the structural, electrical, software, and performance characteristics of TimeCard-based architectures. It describes how hardware subsystems acquire external time references, maintain temporal stability during reference loss (holdover), and accurately distribute disciplined phase and frequency outputs to computing hosts and downstream networks.

The scope of this standardization effort encompasses: - Hardware and software architecture definitions, including unified timescale generation and holdover behavior. - Receive (inbound) and Providing (outbound) timing interface behaviors (e.g., GNSS, PTP, 1PPS, and PCIe PTM). - Out-of-band and in-band management, telemetry, and security mechanisms (e.g., I²C, IPMI, REST). - Quantifiable performance metrics (e.g., ADEV, TDEV, MTIE) and environmental operating constraints. - Structural methodologies for testing and claiming conformance, including functional, performance, and environmental validation procedures based on traceable metrology.

This document is designed to apply equally across: - Vendor-specific commercial mass-production implementations. - Open-source hardware reference designs. - Laboratory, exploratory, or research-grade prototypes.

By defining a **standardized timing subsystem interface**, the specification establishes a foundation for easily interchanging, upgrading across generations, and co-deploying different TimeCards from competing vendors on diverse host platforms while maintaining stringent functional and timing compatibility.

1.2 Purpose

The purpose of this standard is to enable interoperable TimeCard implementations, modular timing subsystem integration, and unambiguous comparison of TimeCard performance claims. It does so by defining common architecture, conformance, interface, control, performance-reporting, environmental, and test-method expectations.

1.3 Motivation

Modern computing workloads increasingly depend on highly precise and deterministic hardware timing. Modern applications—such as **AI model training clusters, high-frequency financial trading, globally distributed Spanner-style databases, and 5G/6G cellular networks**—frequently require sub-microsecond or nanosecond-class synchronization across thousands of independent nodes.

Historically, achieving this level of timing required bespoke, vendor-specific implementations or tightly coupled architectures that lacked modularity and scalability. The **TimeCard architecture** addresses these historical bottlenecks by introducing: - A standardized hardware abstraction layer and signal footprint (e.g., standardized 1PPS, 10 MHz, and Time of Day bounds). - A unified management and telemetry framework supporting diverse transports for configuration and observability. - A common timing distribution mechanism capable of bypassing host operating system latencies using hardware-based timestamping (e.g., PCIe Precision Time Measurement or hardware PTP). - A clear, falsifiable path for cross-vendor conformance certification, grounded in traceable metrology and repeatable test procedures.

The ultimate motivation is to foster an open, competitive ecosystem for precision time distribution.

1.4 Design Philosophy

The P3335 TimeCard specification is founded on several core engineering principles:

Principle	Description
Modularity	The TimeCard operates as a self-contained subsystem that fully abstracts the complexities of oscillator disciplining away from the host CPU.
Interoperability	Management and timing interfaces are standardized to facilitate plug-and-play architectural compatibility across different server fleets and boundary clocks.
Scalability	The architecture gracefully scales from single-node, isolated edge servers to globally synchronized hyperscale network domains, relying on unified ensemble clocks.
Determinism	Hardware-assisted generation and timestamping provide highly predictable, low-jitter, and low-latency outputs, immune to software scheduling variations.
Traceability	Subsystem logic supports an unbroken chain of measurements that reflect recognized international standards (e.g., UTC, TAI), measured with calibrated metrology equipment.
Security and Integrity	Remote management channels and firmware provisioning protocols rely on modern, authenticated, and cryptographically verifiable mechanisms, including data sanitization.

1.5 Relationship to Other Standards

This specification aligns with and periodically references multiple established global timing and measurement standards. Key related standards include:

- **IEEE Std 1588TM-2019 (PTPv2.1)**: Precision Time Protocol for generalized network synchronization.
- **IEEE Std 802.1ASTM-2020**: Timing and Synchronization for Time-Sensitive Applications.
- **ITU-T G.810 / G.8260 / G.8271**: Definitions and boundary parameters for carrier-grade synchronization networks.
- **IEEE Std 1139TM / IEEE Std 1193TM**: Standard definitions and environmental testing methodologies for fundamental frequency metrology quantities (ADEV, TDEV, MTIE).
- **PCI-SIG PCIe Base Specification**: Specifically outlining Precision Time Measurement (PTM) for in-band hardware timestamping.
- **Open Compute Project (OCP) Time Appliances Project (TAP)**: The foundational open-hardware consortium that incubated the original TimeCard concept.

Where applicable, this document references these standards to ensure technical consistency across the industry.

1.6 Structure of the Specification

The standard is organized sequentially into the following major clauses:

Chapter	Description
1. Overview	Provides the motivation, high-level scope, and conceptual structure (Informative).

Chapter	Description
2. Normative References	Lists the foundational standards essential for implementing this specification.
3. Definitions, Acronyms, and Abbreviations	Clarifies exact terminology and abbreviations utilized throughout the text.
4. Conformance	Outlines the formal terminology and methodology for claiming implementation compliance.
5. Architecture	Defines the core TimeCard blueprint, oscillator interactions, and boundary definitions.
6. Performance Specifications	Details the mathematical metrics and methodologies used to quantify stability and accuracy.
7. Timing Interfaces	Specifies the electrical and logical conduits for receiving and providing synchronization.
8. Control Interfaces	Defines the data structures required for out-of-band and in-band management.
9. Environment	Establishes baselines for physical survivability, thermal tolerance, and RF emissions.
10. Applications and Best Practices	Offers structural guidance on real-world deployment and operational optimization (Informative).
Annex A. Metrics	Provides definitions and measurement methodologies for timing and frequency metrics (Informative).
Annex B. Test Procedures	Provides standardized procedures for functional, performance, and environmental testing (Informative).
Annex C. Bibliography	Lists informational references and supplementary reading.

1.7 Intended Audience

This specification is drafted distinctly for: - Hardware, software, and systems engineers developing TimeCard-compatible commercial products or reference designs. - Datacenter architects and system integrators designing synchronized infrastructure and planning for scale-out timing topologies. - Network operators and site-reliability engineers managing latency-sensitive distributed environments who require granular telemetry. - Scientists and researchers requiring high-fidelity timestamping for metrology and specialized physics experimentation. - Test and validation engineers utilize the defined procedures to empirically grade hardware performance and claim conformance. - Affiliated standards bodies (e.g., ITU-T or OCP) seeking structural harmonization with modern end-system timing frameworks.

Readers are expected to possess a foundational understanding of frequency control, phase-locked loops, phase noise characteristics, and packet-based synchronization (e.g., PTP, NTP).

1.8 Strategic Goals

The widespread adoption of the TimeCard architecture aims to achieve the following outcomes: - **Broad vendor-neutral interoperability** among specialized timing payloads and generic host computers, enabling a mature, commercial off-the-shelf (COTS) ecosystem. - **Drastically improved absolute precision** within distributed databases, cellular networks, and distributed AI compute clusters by driving the timescale directly to the hardware bus (e.g., PCIe). - **Lowered integration friction and cost** resulting from strictly defined hardware abstraction interfaces and deterministic holdover behaviors. - **Enhanced operational observability** and traceability to support stringent financial, telecommunications, and governmental regulatory frameworks. - **Consistent metrology evaluations** resulting from standardized, falsifiable performance testing and conformance guidelines.

1.9 Document Status and Origin

This architectural framework draws heavily on the pioneering work in the **Open Compute Project (OCP) Time Appliances Project (TAP)**. Moving this architecture into the formal IEEE P3335 standardization process reflects the industry’s demand for a rigorously governed, universally recognized timing hardware standard.

Contributors, silicon vendors, and system integrators are actively encouraged to submit empirical performance data, interoperability reports, and technical proposals to inform the ongoing evolution of this specification.

2. Normative References

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document, including any amendments or corrigenda, applies.

Informative references and background material are listed in Annex C.

2.1 Standards and Specifications

- **DMTF DSP0236**, *Intelligent Platform Management Interface Specification, Second Generation*, v2.0, 2015.
- **IEEE Std 1139**, *IEEE Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology*.
- **IEEE Std 1193**, *IEEE Guide for Measurement of Environmental Sensitivities of Standard Frequency Generators*.
- **IEEE Std 1588-2019**, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*.
- **IEEE Std 802.1AS-2020**, *IEEE Standard for Local and Metropolitan Area Networks—Timing and Synchronization for Time-Sensitive Applications*.
- **IETF RFC 3411**, *An Architecture for Describing Simple Network Management Protocol (SNMP) Management Frameworks*, 2002.
- **IETF RFC 5905**, *Network Time Protocol Version 4: Protocol and Algorithms Specification*, 2010.
- **IRIG Standard 200-16**, *IRIG Serial Time Code Formats*, 2016.
- **ITU-T Recommendation G.703**, *Physical/electrical characteristics of hierarchical digital interfaces*.
- **ITU-T Recommendation G.810**, *Definitions and terminology for synchronization networks*.
- **ITU-T Recommendation G.8260**, *Definitions and terminology for synchronization in packet networks*.
- **MIPI Alliance Specification for I3C Basic**.
- **PCI-SIG PCI Express Base Specification, Revision 5.0**, 2019.
- **SMBus Forum**, *System Management Bus (SMBus) Specification, Version 3.2*, 2018.

2.2 Reference Use

Normative references are cited in the body of this standard at the point where their content is required to implement an interface, measurement method, or reported characteristic. A referenced document that is used only for examples, background, or deployment guidance is listed in Annex C instead of this clause.

3. Definitions, Acronyms, and Abbreviations

This chapter defines the specialized terms, acronyms, and abbreviations utilized throughout the IEEE P3335 specification.

3.1 Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.

- **Accuracy:** The degree of conformance or closeness of a measured time or frequency value to a defined primary reference (e.g., UTC).
- **Asymmetry Calibration:** The process of calculating and correcting unequal signal propagation delays in bi-directional timing links.
- **Control Plane:** The logical communication path dedicated to configuration, telemetry, and operations management rather than active time distribution.
- **Conformance Statement:** Supplier documentation identifying the IEEE P3335 clauses, optional features, interfaces, environmental limits, and performance declarations for which an implementation claims conformance.
- **Data Plane:** The logical or physical communication path strictly responsible for carrying phase, frequency, and time signals to or from the host or network.
- **Data Plane Latency:** The deterministic or variable propagation delay between a time source and its destination hardware.
- **Deterministic Behavior:** A system characteristic where the timing response under specified operational conditions is highly predictable and repeatable.
- **Disciplining:** The continuous process of steering a local oscillator's frequency and phase to align with a superior external reference signal.
- **Ensemble Clock:** A highly stable composite time source created by mathematically combining multiple independent references via weighting or consensus algorithms.
- **Environmental Chamber:** A specialized testing enclosure used to strictly control ambient temperature, humidity, and atmospheric conditions for hardware thermal testing.
- **Frequency Counter:** A measurement instrument used to determine the exact signal frequency or period over a defined averaging interval.
- **Granularity:** The minimum distinguishable or settable unit of change in a time-domain or frequency-domain output.
- **Hitless Switching:** The ability of a timing device to execute a reference failover or changeover without inducing a measurable phase discontinuity to the downstream consumers.
- **Holdover:** A mode of operation where a timing subsystem continues to maintain accurate time based solely on the historical stability of its internal oscillator after losing its external reference.
- **Holdover Error:** The progressively increasing deviation or drift of the subsystem's timescale from an ideal reference while operating in a holdover state.
- **Host System:** The overarching computing, telecommunications, or networking platform that integrates a TimeCard through a standardized interface boundary.
- **Loop Bandwidth:** The effective frequency range over which a disciplining Phase-Locked Loop (PLL) tracks the reference, determining how fast the loop responds to changes.
- **Lock Time:** The duration required for the subsystem to acquire a reference and assert phase/frequency lock after a cold startup or reference transition.
- **Management Interface:** The designated control channel (e.g., SMBus, IPMI, REST) used strictly for configuration, diagnostics, and firmware management.
- **Measurement Point:** The physical connector, logical interface, register, packet timestamp, or other defined boundary at which a performance value is measured or declared.
- **Optional Feature:** A feature, interface, protocol, environmental profile, or security function that is not required for base conformance but that becomes subject to all applicable mandatory requirements when claimed by an implementation.
- **Oscilloscope (TIE Mode):** Measurement equipment configured to capture the Time Interval Error

(TIE) of repetitive high-frequency signals.

- **Phase Alignment:** The absolute difference in timing between the corresponding edges of two separate signals; typically measured in nanoseconds or picoseconds.
 - **Power Analyzer:** An instrument deployed to measure precise current draw, power sequencing, and consumption events over time.
 - **Precision:** The statistical repeatability, resolution, or variance of a measurement under identical operating conditions.
 - **Primary Reference Clock:** The highest-stability, autonomous time or frequency source in a synchronization hierarchy (as defined by ITU-T G.811).
 - **Providing Interface:** The outbound physical or logical mechanism through which the TimeCard distributes precise time, phase, and frequency representations to the host or downstream devices.
 - **Receive Interface:** The inbound physical or logical mechanism through which the TimeCard acquires an external reference representation.
 - **Reference Signal:** Any external, traceable timing input actively utilized to discipline the local oscillator.
 - **Resolution:** The absolute smallest incremental step, quantization, or granularity of a time or frequency measurement that the system can distinguish.
 - **Rubidium Oscillator:** A highly stable atomic oscillator utilizing rubidium-87 vapor transitions to provide exceptional long-term frequency stability.
 - **TimeCard:** A modular, hardware-based timing subsystem that abstracts synchronization complexity to deliver standardized phase, frequency, and time services to a host system.
 - **Time Interval Counter (TIC):** A highly precise laboratory instrument designed to measure time differences between distinct electrical signal edges with picosecond-level resolution.
 - **Time Jitter:** The short-term variation or instability of a time-domain signal (such as a 1PPS edge) from its ideal position, frequently reported as an RMS or peak-to-peak value.
 - **Traceability:** A documented, unbroken mathematical chain of calibrations linking a local measurement back to recognized primary international standards (e.g., UTC).
 - **Unified Timescale:** A single internal temporal scale actively maintained by the TimeCard, from which all separate outbound interfaces derive coherent, phase-aligned realizations.
 - **Unspecified:** A characteristic intentionally not assigned a universal value by this standard. An unspecified characteristic may still be required to be declared by the supplier for a particular implementation or optional feature.
-

3.2 Acronyms and Abbreviations

- **1PPS:** One Pulse Per Second
- **ADEV:** Allan Deviation
- **API:** Application Programming Interface
- **ASIC:** Application Specific Integrated Circuit
- **BC:** Boundary Clock
- **BMC:** Baseboard Management Controller
- **CDC:** Communications Device Class
- **CXL:** Compute Express Link
- **CSAC:** Chip-Scale Atomic Clock
- **DCLS:** Direct Current Level Shift
- **DDS:** Direct Digital Synthesizer
- **EMC:** Electromagnetic Compatibility
- **EMI:** Electromagnetic Interference
- **ESD:** Electrostatic Discharge
- **FPGA:** Field Programmable Gate Array
- **GNSS:** Global Navigation Satellite System (e.g., GPS, Galileo, GLONASS)
- **GM:** Grandmaster
- **gRPC:** Google Remote Procedure Call

- **GPS:** Global Positioning System
- **I2C:** Inter-Integrated Circuit
- **I3C:** Improved Inter-Integrated Circuit
- **IPMI:** Intelligent Platform Management Interface
- **IRIG:** Inter-Range Instrumentation Group
- **IRNSS:** Indian Regional Navigation Satellite System
- **MAC:** Media Access Control
- **MMIO:** Memory-Mapped Input/Output
- **MTBF:** Mean Time Between Failures
- **MTIE:** Maximum Time Interval Error
- **NC-SI:** Network Controller Sideband Interface
- **NTP:** Network Time Protocol
- **OCF TAP:** Open Compute Project Time Appliances Project
- **OCXO:** Oven-Controlled Crystal Oscillator
- **OC:** Ordinary Clock
- **PCIe:** Peripheral Component Interconnect Express
- **PHC:** PTP Hardware Clock or Physical Hardware Clock
- **PHY:** Physical Layer
- **PLL:** Phase-Locked Loop
- **PN:** Phase Noise
- **PTM:** Precision Time Measurement
- **PTP:** Precision Time Protocol
- **QZSS:** Quasi-Zenith Satellite System
- **REST:** Representational State Transfer
- **RoHS:** Restriction of Hazardous Substances
- **SCPI:** Standard Commands for Programmable Instruments
- **SEC:** Securities and Exchange Commission
- **SMA:** SubMiniature version A (connector)
- **SMBus:** System Management Bus
- **SNMP:** Simple Network Management Protocol
- **SFP:** Small Form-factor Pluggable
- **SWaP-C:** Size, Weight, Power, and Cost
- **TC:** Transparent Clock
- **TAI:** International Atomic Time
- **TCXO:** Temperature-Compensated Crystal Oscillator
- **TDEV:** Time Deviation
- **TIC:** Time Interval Counter
- **TIE:** Time Interval Error
- **TPM:** Trusted Platform Module
- **ToD:** Time of Day
- **USB:** Universal Serial Bus
- **UTC:** Coordinated Universal Time
- **WiWi:** Wireless two-Way interferometry
- **WR:** White Rabbit
- **WWVB:** Radio Station WWVB (NIST)

End of Chapter – Definitions, Acronyms, and Abbreviations

4. Conformance

This clause defines the conditions under which an implementation can claim conformance to IEEE P3335. A conforming implementation shall satisfy the mandatory requirements in this clause and the mandatory

requirements in Clauses 5 through 9 that apply to the implemented features, interfaces, and operating conditions.

Informative clauses, informative notes, and informative annexes provide explanatory material only. They do not create conformance requirements unless a normative clause explicitly references a requirement stated elsewhere.

4.1 Requirement Terms

The following terms are used as defined by IEEE SA rules for standards text:

- **shall** indicates a mandatory requirement.
- **should** indicates a recommended practice among several permissible possibilities.
- **may** indicates a course of action permitted within the limits of this standard.
- **can** indicates capability, possibility, or a statement of fact.

The term **must** is not used as a requirement term in this standard. Where a physical or mathematical inevitability needs to be described, the text should be written as a statement of fact using **is**, **are**, or **can**.

4.2 Conformance Model

IEEE P3335 uses interspersed requirements with a conformance summary. Mandatory and optional requirements are stated in the technical clauses where the relevant behavior is specified. This clause summarizes how those requirements are applied.

A TimeCard implementation may be realized as a discrete add-in card, an embedded module, an FPGA or ASIC function, a system-on-chip block, or an external timing unit. The form factor does not determine conformance. Conformance is determined by the externally observable interfaces, behavior, documentation, and performance declarations of the implementation.

A claim of conformance shall identify:

- The implementation form factor and host interface mapping.
- The timing receive and providing interfaces implemented.
- The control interfaces implemented.
- The optional feature sets claimed.
- The declared operating environment and environmental limits.
- The performance metrics, measurement points, measurement methods, and limits declared by the manufacturer.

4.3 Base Conformance Requirements

A conforming TimeCard implementation shall meet all of the following base requirements:

- The implementation shall provide at least one timing providing interface that distributes time, phase, frequency, or a combination thereof to a host system or downstream consumer.
- All implemented timing providing interfaces shall derive from one unified timescale as specified in Clause 5.
- The implementation shall provide at least one accessible control interface as specified in Clause 8.
- The implementation shall document all implemented timing interfaces, control interfaces, physical connectors, protocol mappings, and optional features.
- The implementation shall document the performance metrics required by Clause 6 for each applicable measurement point.
- The implementation shall document the environmental limits and reliability information required by Clause 9 for the declared deployment environment.
- The implementation shall define the conditions under which each declared performance value is valid, including warm-up time, lock time, reference source, temperature range, and measurement bandwidth where applicable.

- The implementation shall not claim support for an optional feature unless all mandatory requirements associated with that feature are satisfied.

An implementation may conform to IEEE P3335 without implementing every optional receive interface, providing interface, host interface mapping, management protocol, environmental profile, or security hardening feature described in this standard.

4.4 Optional Feature Claims

Optional features are conditional. If an implementation claims support for an optional feature, the implementation shall satisfy the mandatory requirements that apply to that feature.

Optional feature claims include, but are not limited to:

- Receive interfaces such as GNSS, PTP, NTP, White Rabbit, WiWi, WWVB, IRIG, PPS, or frequency-reference inputs.
- Providing interfaces such as PPS, frequency outputs, Time of Day outputs, PTP, PCIe PTM, or other host-bus time-transfer mechanisms.
- Host interface mappings such as PCIe, USB, serial, embedded memory-mapped interfaces, or implementation-specific mappings.
- Control protocols such as SMBus, I2C, I3C, IPMI, NC-SI, REST, gRPC, SNMP, Redfish, serial, or USB.
- Security functions such as authenticated management, signed firmware update, secure boot, encrypted telemetry, key storage, sanitization, or tamper response.
- Environmental profiles such as data-center, telecommunications, industrial, laboratory, or field deployment.
- Ensemble reference processing, holdover classes, or other enhanced timing functions.

Each optional feature claim shall identify the clause or subclause that defines the claimed behavior and shall identify any deviations or implementation-specific limits allowed by that clause.

4.5 Conformance Statement

The supplier of a conforming implementation shall provide a conformance statement. The conformance statement shall be publicly available or supplied with the product documentation and shall include, at a minimum:

- Product name, hardware revision, firmware revision, and relevant configuration profile.
- Claimed IEEE P3335 conformance scope.
- Supported timing receive interfaces and providing interfaces.
- Supported host interface mapping and control interface classes.
- Supported optional feature sets.
- Declared performance metrics and measurement points.
- Declared environmental operating and storage limits.
- References to datasheets, register maps, interface descriptions, calibration information, and test reports needed to evaluate the claim.

The conformance statement should be written so that an independent test laboratory or system integrator can determine which requirements apply without relying on unpublished implementation details.

4.6 Test Evidence

Conformance can be evaluated by functional testing, performance testing, documentation review, or a combination thereof.

Functional testing verifies that the implemented interfaces, state transitions, control operations, and fault responses behave as specified. Performance testing verifies that declared timing, frequency, holdover, and

environmental metrics meet the supplier’s stated limits using the measurement methods declared for the implementation.

Test evidence should identify:

- The device under test and configuration under test.
- The applicable IEEE P3335 clauses and optional feature claims.
- Test equipment, calibration status, and traceability path.
- Test environment, reference source, cabling, and measurement points.
- Pass/fail criteria and measured results.
- Any deviations, waivers, or limitations of the test method.

Informative Annex B provides example test procedures that can be used as a starting point for such evidence.

4.7 Protocol Implementation Conformance Statement

A Protocol Implementation Conformance Statement (PICS) is not defined in this draft. The working group should add a PICS only after the normative requirements are stable enough to avoid divergence between the PICS and the body of the standard.

If a PICS is added in a later draft, the normative clauses shall remain the controlling source of conformance requirements.

5. Architecture (Normative)

This clause defines the architectural requirements for a TimeCard. The architecture is specified in terms of externally observable functions, interfaces, behavior, and documentation. A TimeCard may be implemented as a discrete card, an embedded hardware block, an FPGA or ASIC function, an external module, or another implementation form that satisfies the requirements of this standard.

The architecture separates timing data-plane functions from management and control functions. Timing data-plane functions receive, generate, maintain, timestamp, and provide time, phase, and frequency. Management and control functions configure, monitor, secure, and report the state of those timing functions.

5.1 Architectural Overview

A TimeCard is a timing subsystem that provides time, phase, frequency, or a combination thereof to a host system or downstream consumer. The host system may be a server, telecommunications platform, industrial controller, scientific instrument, embedded system, or other system that requires a traceable timing service.

A conforming TimeCard shall include the following externally observable capabilities:

- A local timing function capable of maintaining a unified timescale.
- At least one timing providing interface.
- At least one control interface.
- Published performance, interface, environmental, and conformance information sufficient for an implementer or operator to evaluate the claimed behavior.

A TimeCard may include one or more timing receive interfaces, one or more time-transfer interfaces, one or more host interface mappings, and one or more reference-selection or ensemble algorithms.

5.2 Implementation Boundary

The TimeCard implementation boundary is the set of interfaces through which the TimeCard exchanges timing signals, timestamp values, control information, status, telemetry, firmware, or power with the host system or external equipment.

An implementation shall document its implementation boundary. The documentation shall identify:

- Host interface mappings.
- Timing receive interfaces, if any.
- Timing providing interfaces.
- Control interfaces.
- Power, thermal, and mechanical dependencies.
- Measurement points used for declared performance values.

Internal partitioning is implementation-specific unless a claimed optional feature or interface mapping makes that partitioning externally observable.

5.3 Core Functional Blocks

A TimeCard architecture can be described by the following logical blocks. An implementation may combine or subdivide these blocks, provided that the externally observable behavior remains conformant.

5.3.1 Local Timing Function

The local timing function maintains the TimeCard’s time, phase, and frequency state. It may be based on a quartz oscillator, OCXO, TCXO, CSAC, rubidium oscillator, atomic reference, DDS, host-provided clock, or another timing source.

The implementation shall document the oscillator or timing-source class used for each declared performance profile. The implementation shall document whether the local timing function is steerable, free-running, disciplined by external references, or operated as a primary reference.

5.3.2 Time Generator

The time generator produces the TimeCard’s unified timescale. It may include counters, phase accumulators, timestamp interpolation, leap-second or timescale conversion logic, frequency steering, phase steering, or other timing logic.

The time generator shall support timestamp representations that permit unambiguous duration computation over the declared operating interval. If an implementation exposes a timescale that is not TAI, UTC, or the IEEE 1588 PTP timescale, the implementation shall document the mapping to the declared reference timescale.

The implementation shall document the supported timestamp range, epoch, granularity, resolution, and behavior at discontinuities such as leap seconds, leap smearing, holdover entry, reference failover, or manual time steps.

5.3.3 Timestamping Facility

If an implementation timestamps events, packets, register accesses, signal edges, or host-bus transactions, the timestamping facility shall derive those timestamps from the unified timescale. The implementation shall document the timestamping measurement point, timestamp latency or correction model, resolution, granularity, and known fixed offsets.

Hardware timestamping should be used for receive and providing interfaces where the supported protocol or physical interface permits it.

5.3.4 Receive Interfaces

Receive interfaces acquire external references that may be used to steer, discipline, initialize, monitor, or validate the unified timescale. Receive interfaces are specified in Clause 7.

An implementation may operate without a receive interface if it is designed for free-running, primary-reference, laboratory, or isolated holdover operation. If no receive interface is implemented, the conformance statement shall state that no external timing reference is supported.

5.3.5 Providing Interfaces

Providing interfaces distribute time, phase, frequency, or timestamp information from the TimeCard to a host system or downstream consumer. Providing interfaces are specified in Clause 7.

At least one providing interface shall be implemented. All providing interfaces implemented by a single TimeCard shall derive from the same unified timescale.

5.3.6 Control Interfaces

Control interfaces provide configuration, monitoring, alarms, telemetry, firmware management, and security operations. Control interfaces are specified in Clause 8.

At least one control interface shall be implemented. A control interface may be local, host-integrated, out-of-band, remote, or a combination thereof.

5.4 Unified Timescale

A TimeCard shall generate one unified timescale for each conforming TimeCard instance. All providing interfaces of that instance shall publish, encode, or derive from that unified timescale.

The implementation shall document:

- The reference timescale used internally or exposed externally.
- The relationship between the unified timescale and UTC, TAI, the IEEE 1588 PTP timescale, or another declared timescale.
- The alignment tolerance among all providing interfaces that claim phase or time alignment.
- The behavior during startup, warm-up, lock acquisition, reference failover, holdover entry, holdover exit, and manual time adjustment.

Analog or periodic providing interfaces that claim continuous phase shall not introduce an undocumented phase step during normal locked operation. If a phase step, holdover event, failover transient, or other discontinuity can occur, its detection and reporting mechanism shall be documented.

5.5 Reference Selection and Ensemble Operation

If more than one receive reference is implemented, the TimeCard shall provide a documented reference-selection policy. The policy shall identify priority, validity, quality, failure detection, fallback, recovery, and operator override behavior.

If an implementation claims ensemble reference operation, it shall document:

- Supported input reference types.
- Weighting, voting, or selection method at a level sufficient for operational evaluation.
- Health and alarm criteria.
- Behavior when one or more ensemble sources degrade or fail.
- Metrics exposed through the control interface.

Ensemble operation shall produce one selected or synthesized reference for disciplining the unified timescale. It shall not create multiple independent conforming timescales within the same TimeCard instance unless those instances are separately identified and separately documented.

5.6 Host and Time-Transfer Interfaces

A host interface mapping defines how a host system discovers, configures, reads, writes, timestamps, or receives time from a TimeCard. PCIe, USB, serial, memory-mapped embedded interfaces, network interfaces, and implementation-specific mappings may be used.

Each host interface mapping shall document:

- Physical and logical interface standards used.
- Discovery and enumeration behavior.
- Required driver-visible registers, messages, commands, or API elements.
- Endianness, alignment, units, and timestamp formats.
- Interrupts, polling requirements, and error reporting.
- Latency, asymmetry, and correction information required for time-transfer evaluation.

If PCIe PTM is implemented, the implementation shall document the PTM capability, the measurement point represented by PTM timestamps, and any correction terms needed to relate PTM time to other providing interfaces.

5.7 Performance and Measurement Architecture

Performance requirements in this standard are expressed as reporting and measurement requirements unless a technical clause explicitly states a numeric limit. This allows TimeCards to serve different use cases while making supplier claims comparable.

An implementation shall identify the measurement point for each declared metric. Metrics shall be reported with enough context to reproduce or evaluate the claim, including reference source, measurement bandwidth, observation interval, environmental conditions, warm-up conditions, and lock state.

The following performance categories shall be addressed when applicable to the implemented interfaces:

- Time accuracy relative to a declared reference.
- Time stability, including MTIE and TDEV where applicable.
- Frequency stability, including ADEV where applicable.
- Phase noise for periodic frequency outputs where applicable.
- Time jitter for pulse outputs where applicable.
- Holdover error versus elapsed time.
- Lock acquisition, recovery, and failover behavior.
- Environmental sensitivity relevant to the declared operating profile.

Clause 6 defines the required reporting details for these categories. Annex A provides informative metric background.

5.8 Power, Mechanical, and Environmental Architecture

Physical TimeCard implementations shall document the power, mechanical, thermal, connector, and environmental assumptions needed for safe operation and declared timing performance.

Embedded or virtualized TimeCard implementations shall document the host resources, clock dependencies, thermal assumptions, and integration constraints that affect declared timing performance.

Detailed environmental requirements are specified in Clause 9.

5.9 Optional Features and Extensions

An implementation may include optional features or vendor-specific extensions. Optional features and extensions shall not prevent baseline operation of the documented conforming interfaces.

Vendor-specific extensions shall be documented sufficiently to allow an operator to distinguish standardized P3335 behavior from implementation-specific behavior. Reserved values, register fields, message types, and identifiers shall not be repurposed in a way that conflicts with this standard.

5.10 Documentation Requirements

Manufacturers shall provide documentation sufficient to evaluate and integrate a conforming TimeCard. The documentation shall include:

- Conformance statement required by Clause 4.
- Interface descriptions for all receive, providing, host, and control interfaces.
- Register maps, information models, schemas, command descriptions, or API descriptions needed to use the implemented control interfaces.
- PLL or disciplining-loop type and bandwidth, or a statement that the implementation does not use such a loop.
- Warm-up time to first usable output and warm-up time to declared full performance.
- Lock acquisition time, failover behavior, holdover entry and exit behavior, and recovery behavior.
- Declared performance metrics and measurement points required by Clause 6.
- Environmental and reliability information required by Clause 9.
- Traceability information sufficient for operational analysis, or an explicit statement that traceability is not supported.
- All optional features claimed and all conditional requirements exercised by those claims.

Documentation should be publicly available for commercial products and available to integrators for non-commercial, embedded, or custom implementations.

5.11 Informative Time-Flow Narrative

In a typical TimeCard, one or more receive interfaces acquire external timing references such as GNSS, PTP, PPS, frequency references, or other sources. Reference-selection logic evaluates those inputs and selects or synthesizes a reference. A disciplining function steers the local timing function. The time generator maintains the unified timescale. Providing interfaces distribute that unified timescale as PPS, frequency, Time of Day, packet timestamps, host-bus time, or other representations. Control interfaces report status, telemetry, alarms, and configuration state.

When all usable external references are lost, the TimeCard normally enters holdover or free-running operation. During holdover, the unified timescale continues according to the local timing function and any holdover model implemented by the TimeCard. The TimeCard reports the state transition and exposes the information needed to evaluate the declared holdover behavior.

6. Performance Specifications (Normative)

This clause defines the performance reporting and characterization requirements for TimeCard implementations. The standard does not define a single universal numeric performance class for all use cases. Instead, it requires performance claims to be stated using common metrics, defined measurement points, traceable methods, and declared operating conditions.

6.1 Performance Declaration Model

Manufacturers shall publish performance declarations for each conforming TimeCard implementation. Each declared value shall identify:

- The metric being reported.
- The measurement point.
- The operating mode, such as locked, holdover, free-running, warm-up, failover, or recovery.
- The reference source or reference timescale.
- The applicable temperature range, voltage range, airflow or cooling condition, and other environmental conditions.
- The minimum warm-up time and lock time required before the declaration applies.
- The observation interval, averaging time, bandwidth, sample count, or statistical confidence basis used.
- The measurement equipment class and traceability path.

Performance values should be stated as bounded claims rather than typical-only values. Typical values may be provided in addition to bounded claims.

6.2 General Measurement Requirements

All measurements used to support conformance claims shall be made with equipment traceable to a recognized national metrology institute or other declared primary reference. Measurement methods shall conform to the applicable definitions in IEEE Std 1139, IEEE Std 1193, ITU-T G.810, or ITU-T G.8260 when those metrics are used.

The measurement uncertainty, instrument noise floor, measurement bandwidth, averaging configuration, and environmental conditions shall be documented with the reported result. The measurement point shall be identified in enough detail that another laboratory can reproduce the setup.

If the term **jitter** is used, the implementation documentation shall define the exact jitter metric, measurement bandwidth, sample population, and statistical calculation. Time jitter, phase noise, MTIE, TDEV, and ADEV shall not be used interchangeably.

6.3 Time Accuracy

Time accuracy declarations shall state the error of a TimeCard output or timestamp relative to a declared reference timescale or source.

For each time accuracy declaration, the manufacturer shall document:

- Source of standard time, such as GNSS system time, UTC(k), TAI, PTP grandmaster time, or another declared reference.
- Measurement point, such as 1PPS output, Time of Day output, PTP egress timestamp, PTM timestamp, or register-read timestamp.
- Maximum time error or another explicitly defined bounded statistic.
- Statistical basis, such as maximum observed value, percentile, RMS value, or confidence interval.
- Static error and dynamic error components when they are separately known.
- Valid temperature range and environmental profile.
- Minimum lock time before the declaration applies.
- Relationship between the declared source and UTC or TAI, if the declared source is not itself UTC or TAI.

If an implementation reports IEEE 1588 clockAccuracy values, the values shall correspond to the measured or declared accuracy range for the applicable operating mode.

6.4 Time Stability

Time stability declarations shall characterize variation of the TimeCard time output or timestamp stream over a defined observation interval.

Manufacturers shall report MTIE for time outputs or timestamp streams for which bounded time-error behavior is claimed. The MTIE declaration shall include observation intervals, measurement point, lock state, environmental conditions, and applicable limit or measured result.

Manufacturers should report TDEV for time outputs or timestamp streams where stochastic time deviation is operationally relevant. TDEV declarations shall include averaging intervals and the method used to compute the result.

6.5 Frequency Stability and Phase Noise

Frequency stability declarations shall characterize the frequency output or local timing function over one or more averaging intervals.

Manufacturers shall report frequency accuracy and temperature stability for the local timing function or for each frequency output for which a frequency-performance claim is made.

Manufacturers should report ADEV for frequency outputs or oscillator functions where long-term or short-term frequency stability is relevant. ADEV declarations shall include the averaging intervals and measurement point.

For periodic outputs such as 10 MHz, manufacturers should report phase noise as a function of offset frequency. Phase-noise declarations shall identify carrier frequency, offset-frequency range, measurement bandwidth, instrument configuration, and measurement point.

6.6 Pulse and Event Timing

For pulse outputs such as 1PPS, manufacturers shall document:

- On-time edge definition.
- Output polarity.
- Pulse width range.
- Rise and fall time measurement method.
- RMS time jitter or another explicitly defined short-term timing variation metric.
- Peak-to-peak or bounded time variation when claimed.
- Alignment to other providing interfaces of the same TimeCard.

For event-capture or timestamping inputs, manufacturers shall document the event measurement point, timestamp resolution, granularity, latency, fixed corrections, and known uncertainty contributors.

6.7 Holdover Performance

Holdover declarations shall characterize accumulated time error after loss of all applicable external synchronization references.

For each holdover declaration, the manufacturer shall document:

- Holdover entry condition and triggering event.
- Reference source and lock duration before holdover begins.
- Minimum warm-up and stabilization conditions before holdover begins.
- Measurement point.
- Maximum time error or MTIE versus elapsed holdover time.
- Temperature profile and voltage conditions during holdover.
- Behavior when the reference is restored.

MTIE as defined by ITU-T G.810 or ITU-T G.8260 shall be used when reporting bounded holdover time error. Other holdover indicators, such as aging rate or estimated drift, may be reported in addition to MTIE.

6.8 Dynamic Operation

Manufacturers shall characterize timing behavior during operational transitions that can affect output phase, frequency, or timestamp values. The characterization shall include, where applicable:

- Cold-start and warm-start lock acquisition time.
- Transition from locked operation to holdover.
- Transition from holdover to locked operation.
- Reference failover between two valid sources.
- Manual time step or frequency steering command.
- Firmware update or restart behavior if timing service is interrupted or degraded.

For each transition, the manufacturer shall document whether the unified timescale remains continuous, whether an output phase step can occur, and how the condition is reported through the control interface.

6.9 Environmental and System-Integration Effects

Performance declarations shall identify the environmental and host-system conditions under which the declared values apply. The following effects shall be characterized when applicable to the declared deployment profile:

- Temperature and thermal-gradient sensitivity.
- Input-voltage variation and supply noise sensitivity.
- Vibration sensitivity.
- Airflow and host thermal loading.
- Host-bus activity, electromagnetic interference, or crosstalk that can affect timing outputs.

If performance is declared only for a subset of the environmental range specified in Clause 9, that subset shall be explicitly stated.

6.10 Performance Documentation Checklist

The performance section of a datasheet or conformance statement shall include the following items when applicable:

Category	Required reporting content
Time accuracy	Reference source, measurement point, bounded error, statistic, temperature range, lock time
Time stability	MTIE and applicable TDEV intervals, measurement point, state, environmental profile
Frequency stability	Frequency accuracy, temperature stability, applicable ADEV intervals, measurement point
Phase noise	Carrier frequency, offset-frequency range, measurement bandwidth, phase-noise curve or limits
Pulse timing	Edge definition, pulse width, rise/fall time, time jitter definition, alignment to unified timescale
Holdover	Entry condition, prior lock duration, MTIE/error versus elapsed time, temperature profile
Transitions	Lock acquisition, failover, holdover exit, phase steps, reporting mechanism
Environment	Temperature, voltage, vibration, airflow, and host-integration assumptions

7. Timing Interfaces (Normative)

This chapter defines the physical, electrical, and logical interfaces through which a TimeCard exchanges timing and frequency information with external systems. These interfaces form the foundation of interoperability between diverse TimeCards, host platforms, and external timing networks. All implementations shall adhere to the standardized behaviors defined herein to facilitate cross-vendor compatibility and predictable operation.

7.1 Overview

A TimeCard typically incorporates three fundamental classes of interfaces: 1. **Receive (Inbound) Interfaces:** Dedicated to acquiring timing signals from external authoritative references. 2. **Providing (Outbound) Interfaces:** Dedicated to distributing disciplined time, phase, and frequency representations to the host system and other downstream consumers. 3. **Management and Control Interfaces:** Non-timing-critical data paths used for configuration, telemetry, security, and diagnostics.

Each class of interface has distinct electrical, signaling, protocol, and performance requirements, detailed in the following subclauses.

7.2 Receive (Inbound) Interfaces

The receive interface enables the TimeCard to lock its internal oscillator function to one or more external references, establishing traceability and synchronization to a higher-order time source.

7.2.1 Supported Reference Types

The following external references are recognized for TimeCard synchronization. A TimeCard may support one or more of these inputs: - **GNSS Receivers:** (e.g., GPS, Galileo, GLONASS, BeiDou) supplying 1 Pulse Per Second (1PPS) and Time of Day (ToD) streams such as NMEA or binary protocols. - **Precision Time Protocol (PTP):** Packet-based synchronization via IEEE Std 1588 over Ethernet or PCIe networks. If implemented, PTP protocol behavior shall conform to IEEE Std 1588-2019 or IEEE Std 802.1AS-2020 for the claimed profile. - **Network Time Protocol (NTP):** Packet-based synchronization for administrative or lower-accuracy time distribution. If implemented, NTP protocol behavior shall conform to IETF RFC 5905. - **White Rabbit (WR):** Sub-nanosecond synchronization via deterministic Ethernet variants. - **WiWi:** Wireless precision timing leveraging IEEE 802.1AS extensions. - **Regional Radio Standards:** (e.g., WWVB, DCF77, MSF) for localized low-frequency time distribution. - **Direct Electrical References:** 1PPS and continuous frequency (e.g., 10 MHz) continuous-wave or square-wave inputs for direct discrete phase locking or localized chaining. - **IRIG Time Codes:** AM or DCLS time-code inputs. If implemented, IRIG time-code inputs shall conform to IRIG Standard 200-16 for the claimed format.

7.2.2 Interface Characteristics

For each physical receive interface implemented, the manufacturer shall specify the following characteristics in the product documentation: - **Electrical level:** (e.g., 3.3V CMOS, LVTTTL, RS-422, sinewave dBm). - **Input impedance and line termination:** (e.g., 50 Ω , 75 Ω , high-Z). - **Signal polarity and edge definition:** For pulse inputs, defining whether the rising or falling edge acts as the on-time phase marker. - **Connector and pin assignment:** Including cable, optical, antenna, or host-bus connector details. - **Timing tolerance:** Including pulse width, rise/fall time, latency, asymmetry, or other parameters relevant to the interface.

Additionally, the lock acquisition time parameters and holdover entry/exit behaviors are configurable properties of the TimeCard and shall be documented.

7.2.3 Input Redundancy and Selection

TimeCards supporting multiple reference inputs shall implement source selection and failover logic (e.g., an implementation of the Best Master Clock Algorithm or a proprietary multi-source weighting ensemble).

The source priority, weighting, and failover policy shall be configurable via the management interface. During a reference transition (failover), the TimeCard internal oscillator should maintain phase continuity; however, derivative outputs are permitted bounded discontinuities as defined by the manufacturer's specification.

7.2.4 Sensitivity and Thresholds

For each physical receive path, the manufacturer shall declare valid input amplitude, impedance, pulse-width, rise/fall-time, and noise-tolerance limits. If a receive interface claims conformance to another standard, the interface shall meet the limits of that referenced standard at the declared measurement point.

Loss-of-signal (LOS) detection boundaries and subsequent signal reacquisition thresholds shall be explicitly documented in the vendor's datasheet.

7.3 Providing (Outbound) Interfaces

The providing interface is responsible for delivering the unified, synchronized timescale generated by the TimeCard to the host platform or external physical devices.

7.3.1 Output Signal Classes

Typical providing interfaces include, but are not limited to: - **Pulse-Per-Second (PPS)**: The primary second-aligned discrete reference pulse. - **Frequency Outputs**: Continuous clocks (e.g., 10 MHz, 5 MHz) used for frequency synchronization. - **Time of Day (ToD)**: Serial-formatted absolute time messages (e.g., NMEA, IRIG-B, or proprietary binary telemetry). - **Packetized Time (PTP)**: IEEE 1588 traffic delivered via Ethernet ports or directly into the host via PCIe networking abstractions. - **In-Bus Timing (PTM)**: PCIe Precision Time Measurement transactions directly delivered to the host bridging architecture. - **Synchronous Clocks**: Discrete phase-aligned clocks for driving localized FPGA or SoC logic domains. - **IRIG Time Codes**: AM or DCLS time-code outputs. If implemented, IRIG time-code outputs shall conform to IRIG Standard 200-16 for the claimed format.

7.3.2 Electrical and Protocol Standards

All physical outputs shall comply with the following baseline signaling levels unless otherwise documented for specialized physical environments: - **PPS and frequency outputs**: 3.3 V CMOS or LVTTTL utilizing 50 Ω source impedance. - **ToD / Serial outputs**: RS-232, RS-422, or RS-485 compliant signaling levels. - **PCIe / PTM interfaces**: Compliant with PCI-SIG PCI Express Base Specification Revision 5.0, or with a later revision explicitly declared in the conformance statement. - **Ethernet / PTP interfaces**: Compliant with IEEE Std 1588-2019 and/or IEEE Std 802.1AS-2020.

For each implemented providing interface, the manufacturer shall document the signal rise/fall times, maximum intrinsic jitter or other declared timing variation metric, source impedance, absolute drive capability, connector, and measurement point.

If a PPS or frequency output claims compliance with ITU-T G.703 synchronous signaling, that output shall conform to the applicable ITU-T G.703 requirements at the declared measurement point.

7.3.3 Synchronization Accuracy

All output signals shall align coherently with the single unified timescale generated natively by the TimeCard, as described in Clause 5.

The manufacturer shall declare the maximum alignment error between each providing interface and the unified timescale. If an implementation claims alignment among multiple providing interfaces, the manufacturer shall declare the measurement points, bounded alignment error, statistic used, and operating conditions. Output clocks should exhibit deterministic phase behavior and bounded frequency deviation during an internal source failover event.

7.3.4 Hardware Timestamping

Providing interfaces shall support hardware-assisted timestamping where technically feasible within the protocol, including: - Direct physical PPS edge capture. - In-band PCIe PTM transaction stamping. - PTP transmit (egress) and receive (ingress) hardware packet timestamps.

Timestamping accuracy validation shall be performed using measurement equipment directly traceable to at least one recognized National Metrology Institute (e.g., NIST, NPL, or PTB).

7.4 Management and Control Interfaces

Management interfaces define the logical communications paths dedicated to device configuration, telemetry polling, and firmware lifecycle control. While these interfaces do not carry the critical synchronization time path, they are indispensable for operational deployment.

7.4.1 Common Management Interfaces

Interface	Typical Use	Relative Bandwidth	Example Use Case
SMBus / I²C	Low-level baseboard configuration	Low	Host BMC telemetry, power sequencing, clock control
IPMI / NC-SI	Out-of-band server management	Medium	Platform health monitoring, remote chassis querying
PCIe Config Space	In-band direct host control	High	Runtime MMIO status registers, PCIe PTM enablement
Serial / USB	Maintenance and physical access	Medium	Localized firmware flashing, debug CLI access
REST / gRPC / SNMP	Remote network orchestration	Variable	Distributed datacenter timing management systems

7.4.2 Management Capabilities

Where a management interface is provided, it shall allow read and write access to at least the following operational vectors: - Active synchronization source selection and current fallback state. - Internal oscillator disciplining mode (e.g., locked, holdover, free-run, warming up). - Phase and frequency offset statistics relative to the reference. - Active alarm registers, event logs, and threshold warnings. - Firmware versioning, hardware revisions, and active security status.

7.4.3 Security and Sanitization Requirements

Given the critical nature of timing infrastructure, management interfaces shall adhere to strict security protocols: - All remote or networked management channels shall require standard authentication mechanisms (e.g., username/password, cryptographic tokens, or mutual TLS certificates). - Firmware payload updates shall mandate cryptographic signing and boot-time integrity verification to prevent unauthorized code execution. - Telemetry data crossing shared or external network boundaries should be encrypted. - **Data Sanitization:** TimeCards should support a cryptographic erase or sanitization procedure. Upon receiving a specific management command or upon detection of a physical tamper event, the system shall permanently erase all sensitive internal state data, private keys, and operational logs when the implementation claims a sanitization feature.

7.5 Physical and Logical Integration

7.5.1 Connectors and Labeling

Physical TimeCard front panels (brackets) shall clearly label all timing and management ports. For example: [GNSS] [PPS OUT] [10MHz] [ToD] [MGMT] [USB].

Manufacturers should adopt standardized color coding and iconography in accordance with Open Compute Project Time Appliance Project (OCP-TAP) guidelines where applicable.

7.5.2 Interoperability Goals and Requirements

- All providing interfaces on a single TimeCard shall publish the same mathematically aligned, unified timescale.
- Interfaces should support broad cross-vendor interoperability by utilizing common electrical and logical formats specified herein.
- While vendors may implement proprietary or optional extensions to the TimeCard interface, such extensions shall not disrupt or degrade baseline interoperability with the standardized interfaces claimed by the implementation.

7.6 Summary

Timing interfaces construct the foundational boundaries of TimeCard interoperability, facilitating precise, deterministic, and hardware-agnostic synchronization. By following the physical and logical interface definitions in this chapter, any TimeCard—regardless of its underlying design or vendor—can integrate into diverse host environments, maintain traceable absolute time accuracy, and distribute stable frequency and phase alignments.

8. Control Interfaces (Normative)

This chapter defines the physical, logical, and protocol-level **control interfaces** used to configure, monitor, and manage TimeCard devices.

8.1 Overview

Control interfaces form the out-of-band and in-band **management plane** of the TimeCard architecture. Unlike the timing interfaces defined in Clause 7, which operate strictly within the data plane to deliver phase and frequency synchronization, control interfaces are responsible for:

- Configuration of synchronization parameters and state machines.
- Status monitoring, health telemetry, and fault reporting.
- Firmware provisioning and lifecycle security management.
- Traceability reporting and calibration operations.

Every compliant TimeCard implementation shall provide at least one accessible control interface. This interface may be exposed through a physical baseboard connector, a host-exposed bus interface, or a network-based protocol suite.

8.2 Control Interface Classes

Control interfaces are categorized based on their communication topology and data granularity.

Class	Description	Typical Use
Local Hardware Control	Direct access via SMBus, I ² C, I3C, or GPIO.	Baseboard-level configuration, power sequencing, or low-level telemetry by a BMC.
Host-Integrated Control	In-band access via PCIe MMIO or USB.	Runtime synchronization control by host OS software or drivers.
Out-of-Band (OOB) Control	NC-SI, IPMI, or dedicated serial interface.	Independent platform management regardless of the host OS state.
Remote Network Control	REST, gRPC, Redfish, or SNMP.	Distributed system orchestration or cloud-scale management.

A TimeCard may implement one or more of these control classes simultaneously to support sophisticated datacenter abstraction models.

8.3 Minimum Functional Requirements

Where a control interface is implemented, it shall expose at least the following minimum common set of capabilities, restricted by the bandwidth and constraints of the specific medium:

Function	Description
Source Selection	Select, prioritize, or configure the active synchronization reference (e.g., GNSS, PTP, PPS).
Mode Control	Read and modify the current operational state machine (e.g., Locked, Holdover, Free-running, Warm-up).
Clock Discipline Settings	Adjust PLL or disciplining loop time constants and bandwidth parameters.
Telemetry Access	Retrieve metrics such as fractional frequency offset, internal temperatures, instantaneous phase error, and stability statistics.
Alarm and Event Logs	Report critical operational events such as loss of reference, spoofing detection, or firmware integrity faults.
Firmware Management	Support cryptographic firmware update triggers, version reporting, and rollback.

8.4 Communication Protocols

8.4.1 SMBus / I²C / I3C

- Provides low-speed, low-latency configuration and sensor access, typically used by a Baseboard Management Controller (BMC).
- The address space and register map should be modeled consistently across compliant implementations to ease integration.
- Bus arbitration, clock stretching, and timing compliance shall follow the respective SMBus or MIPI I3C specifications.

8.4.2 PCIe Configuration and MMIO

- Allows in-band, high-speed access to TimeCard control registers directly from the host CPU.
- The register layout should comply with recognized industry standard memory maps (such as the OCP-TAP Control Register Map) where applicable.
- If the TimeCard supports PCIe Precision Time Measurement (PTM), the control interface shall present the necessary standard capability structures to the host to enable PTM negotiation.

8.4.3 IPMI / NC-SI

- Enables robust out-of-band control independent of the host processor or operating system.
- IPMI implementations shall conform to DMTF DSP0236 for the claimed IPMI version and command scope.
- TimeCard devices establishing IPMI communication should implement an extended command set for time synchronization management (e.g., querying clock state, initiating reference failover).
- IPMI commands may support authenticated sessions utilizing pre-shared platform credentials.

8.4.4 REST / gRPC / SNMP / Redfish

- Used for high-level, network-based management and large-scale telemetry aggregation.
- REST, Redfish, and gRPC APIs should conform to established management schemas where such schemas apply.
- SNMP implementations shall conform to the SNMP architecture defined by IETF RFC 3411 for the claimed SNMP version and management scope.
- SNMP implementations should expose standard OIDs for timing health, synchronization source, and hardware/firmware versions.

8.4.5 Serial and USB

- Serial (RS-232/RS-422/RS-485) interfaces should provide a human-readable command-line shell or SCPI-like syntax for localized debugging.
- USB connections may present as standardized Communications Device Class (CDC) / virtual COM ports to facilitate maintenance or localized firmware provisioning.

8.5 Control Register Structure

The logical control registers listed below represent the fundamental baseline for low-level interaction between the host/BMC and the TimeCard. While physical implementations differ (e.g., I2C addresses vs. PCIe MMIO offsets), the logical concepts shall be present:

Logical Register Concept	Description	Access	Example Units
SYNC_SRC	Active external synchronization source ID	R/W	Enumeration
PLL_STATE	Current disciplining state (Locked, Holdover, Free)	R	Enumeration
PHASE_ERR	Instantaneous phase error relative to the primary reference	R	nanoseconds
TEMP	Active local oscillator temperature	R	°C
MTIE_EST	Current estimated holdover drift/MTIE boundary	R	nanoseconds
FIRMWARE_VER	Active running firmware version string	R	ASCII / Hex
UPDATE_CMD	Trigger flag for firmware update sequences	W	Boolean
SECURE_MODE	Runtime security lockdown indicator	R/W	Boolean

All registers should adhere to consistent endianness and word alignment across a vendor's product line.

Reserved values are reserved for future use by IEEE P3335. A reserved value shall not be configured by an implementation or management client. If a reserved value is read from a field, the management client shall ignore that value unless a later revision of this standard defines it.

8.6 Security and Access Control

Note: The requirements in this subclause apply conditionally. They are required only for TimeCards marketed or designated for secure infrastructure deployments. For isolated or lab-grade physical environments, these features may be omitted to reduce complexity.

Where security hardening is implemented, control interfaces operate as primary attack surfaces and require robust cryptographic protection.

8.6.1 Authentication and Authorization

- Networked control protocols shall support user or machine authentication.

- Privilege levels should distinguish between read-only monitoring access, operator configuration access, and administrator provisioning access.
- Access credentials and cryptographic keys should be stored securely within a hardware-backed enclave or discrete TPM.

8.6.2 Secure Firmware and Configuration

- Firmware payload images shall be digitally signed using verifiable certificates.
- A secure boot sequence should verify firmware integrity (cryptographic hash matches) at initialization before granting control to the TimeCard operating logic.

8.6.3 Network Security

- Remote IP-based interfaces (REST, Redfish, gRPC) should utilize encrypted transport layers (e.g., TLS).
- Replay and injection protection should be implemented for operations traversing shared networking environments.

8.7 Event and Telemetry Management

To facilitate continuous performance observability, the control interface should support structured event and telemetry generation.

8.7.1 Event Reporting

- Generated operational events shall include a local timestamp, a severity classification, and a source identifier.
- Critical events should include, but are not limited to:
 - Reference loss (LOS)
 - Phase lock achieved or lost
 - Holdover entry or exit
 - Temperature threshold alarms
 - Firmware update success or failure.

8.7.2 Telemetry Streaming

- Continuous telemetry may be streamed via IPMI, REST, or gRPC for integration into centralized infrastructure dashboards.
- Telemetry generation rates shall be configurable by the administrator.
- Time synchronization statistics (such as ADEV or instantaneous phase offsets) should be strictly timestamped and aligned to the TimeCard's unified timescale.

8.7.3 Logging and Traceability

- Critical alarm logs should be preserved in non-volatile memory across power cycles.
- Exported log files may include cryptographic signatures to validate log integrity during forensic analysis.
- Traceability metadata (such as calibration constants and firmware build hashes) should be accessible to support operational compliance audits.

8.8 Interoperability and Extensions

All implemented control interfaces shall comply with the baseline schema defined by the overarching IEEE P3335 architecture.

Manufacturers may implement proprietary extensions, custom registers, or vendor-specific telemetry strings. These optional extensions shall not compromise or interfere with the baseline interoperability of the standardized registers and capabilities.

8.9 Summary

Control interfaces provide the framework for secure, consistent, vendor-neutral lifecycle management of TimeCard systems.

By standardizing configuration and telemetry logic, this framework facilitates integrating diverse TimeCards across heterogeneous server fleets and telecommunications infrastructures. This architectural consistency promotes long-term maintainability, traceability, and operational reliability throughout the lifespan of precision time synchronization networks.

End of Chapter – Control Interfaces

9. Environmental Specifications (Normative)

This chapter defines the environmental, mechanical, and reliability parameters applicable to compliant TimeCard implementations. The objective is to establish predictable hardware behavior under diverse deployment conditions—ranging from controlled data centers and telecommunications facilities to industrial or field installations—while maintaining timing precision and conformance to the unified timescale.

9.1 Overview

TimeCards are precision timing subsystems whose performance is directly influenced by environmental factors such as ambient temperature, thermal gradients, humidity, mechanical shock, vibration, and electromagnetic interference.

All implementations shall specify their environmental operating limits, test methodologies, and mitigation measures in compliance with the standards to which the manufacturer claims conformity.

Environmental performance should be validated through qualification testing to verify that frequency, phase, and absolute time accuracy remain stable and within stated bounds over the entire declared operating profile.

9.2 Operating Conditions

9.2.1 Temperature

- **Operating Range:** Each TimeCard shall explicitly define a continuous operating temperature range suited for its target deployment (e.g., 0°C to +55°C for data center use, or −40°C to +85°C for industrial use).
- **Storage Range:** Vendors shall specify safe, non-operational storage temperatures and humidity levels.
- **Thermal Stability:** Local oscillator frequency drift versus temperature (temperature coefficient) shall be characterized and documented.
- **Compensation:** Precision devices employing Oven-Controlled Crystal Oscillators (OCXO) or Temperature-Compensated Crystal Oscillators (TCXO) should document their integrated temperature compensation logic or calibration tables.
- **Thermal Alarms:** Active management telemetry, if present, shall include configurable over-temperature and under-temperature warning thresholds.

9.2.2 Humidity

- **Operating Humidity:** Devices shall be specified to operate safely between 5% to 95% non-condensing relative humidity, unless an alternative range is explicitly declared.
- **Environmental Sealing:** Conformal coating, potting, or physical sealing should be applied for models explicitly marketed for humid, corrosive, or outdoor environments.
- Condensation prevention bounds shall be documented, and survivability under thermal cycling should be verified.

9.2.3 Altitude and Airflow

- Operating altitude range and corresponding thermal derating curves shall be defined (e.g., typically 0–3000 meters above sea level).
 - Host airflow assumptions shall be documented, including the minimum required airflow (e.g., Linear Feet per Minute - LFM or CFM) to guarantee safe heat dissipation as a direct function of the inlet air temperature.
-

9.3 Mechanical and Structural Requirements

9.3.1 Form Factor

- The physical envelope shall be documented and should conform to an industry-standard dimensioning specification (e.g., PCIe low-profile, full-height, OCP NIC 3.0, or standard MEZZ).
- Custom, proprietary, or deeply embedded modules shall explicitly document mounting hole patterns, insertion force limits, and recommended connector retention torque levels.

9.3.2 Shock and Vibration

- TimeCards should withstand mechanical shock and vibration profiles aligned with their intended use cases. *(Note: For example, standard server-grade profiles may utilize IEC 60068-2-6 and IEC 60068-2-27 for vibration and shock, respectively, while aerospace or heavy industrial profiles may dictate stricter constraints. A single shock requirement is not broadly applicable across all TimeCard deployment profiles.)*
- The mechanical isolation of the oscillator from board-level vibration is a matter of proprietary design.
- The manufacturer shall document the maximum frequency stability degradation (e.g., acceleration sensitivity, Γ , measured in parts per g) under anticipated vibration levels.

9.3.3 Connectors and Retention

- RF and timing reference ports shall utilize locking, threaded, or high-retention physical connectors (e.g., SMA, SMB, MCX, MMCX).
 - Faceplate physical labeling shall clearly identify the function of each exposed port (e.g., GNSS, PPS, 10 MHz, ToD, MGMT).
 - Appropriate physical cable strain relief should be incorporated to prevent mechanical fatigue on critical RF solder joints.
-

9.4 Electrical and Power Environment

9.4.1 Power Supply

- TimeCards shall distinctly define input power rail nominal voltages and required absolute tolerances (e.g., +12 V $\pm 5\%$, +3.3 V $\pm 3\%$).
- Devices physically powered via standard host buses (e.g., PCIe slots) shall comply with the bus specifications for power sequencing, inrush current, and maximum continuous current draw limits.

- External power connectors should include localized reverse-polarity and surge protection logic.
- Localized energy storage (e.g., supercapacitors or lithium backup batteries) may be utilized to sustain RTC or holdover operation during brief power outages.

9.4.2 Electromagnetic Compatibility (EMC)

- Implementations shall identify the EMC emission and immunity requirements applicable to the declared destination market (e.g., EN 55032, EN 55035, FCC Part 15 Subpart B) and shall document the compliance basis used for the claim.
- Baseboard shielding, ground-plane flooding, and oscillator enclosure isolation to exclude EMI are matters of proprietary vendor design.

9.4.3 Electrostatic Discharge (ESD)

- ESD protection mechanisms shall be provided on all externally exposed connectors in accordance with relevant electrical standards (e.g., IEC 61000-4-2 limits for contact and air discharge).
- Safe handling procedures and static warning labels should be included in both manufacturing integration documentation and end-user manuals.

9.5 Environmental Qualification Tests

All TimeCards intended for commercial production should undergo a formalized qualification testing matrix. Vendors may utilize the following baseline parameters, or add validated procedures specific to their target market:

Test Profile	Reference Standard	Primary Purpose
Thermal Cycling	IEC 60068-2-14	Validate oscillator phase stability across temperature extremes.
Humidity Endurance	IEC 60068-2-78	Assess component corrosion limits and moisture protection.
Mechanical Shock	IEC 60068-2-27	Verify PCB fracture survivability and mechanical retention.
Random Vibration	IEC 60068-2-64	Characterize phase noise and frequency stability under active vibration.
ESD Immunity	IEC 61000-4-2	Verify localized protection against electrostatic discharge.
EMC Emission/Immunity	EN 55032 / EN 55035	Baseline regulatory compliance.
Power Interruption	IEC 61000-4-11	Evaluate state-machine restart and holdover recovery transitions.

Performance deviation limits recorded during post-qualification testing shall not exceed the vendor's formally declared tolerances.

9.6 Reliability and Lifetime

9.6.1 Mean Time Between Failures (MTBF)

- Vendors shall specify an expected MTBF utilizing recognized reliability modeling standard (e.g., Telcordia SR-332, MIL-HDBK-217F, or field-measured historical data).

- MTBF goals should be scaled to the target environment (e.g., >100,000 hours for standard datacenters, or >50,000 hours for harsh field conditions).

9.6.2 Wear-Out and Service Life

- Hardware components susceptible to finite lifespans or wear-out (e.g., lithium backup batteries, localized cooling fans, flash memory write cycles) shall distinctly list their rated service intervals in the product documentation.
- Essential internal device configuration and persistent calibration data shall safely persist across expected power cycles over the device's operational lifetime.

9.6.3 Calibration and Traceability

- TimeCards should undergo individual factory calibration against a traceable primary standard (e.g., UTC(NPL), UTC(NIST)) before delivery.
- Available calibration certificates should record the date of calibration, the strict environmental conditions present during the test, and the calculated margins of uncertainty.

9.7 Supplemental Compliance and Documentation Guidance (Informative)

The material in this subclause is informative. Hardware vendors remain responsible for the regulatory constraints, safety laws, and environmental policies of the jurisdictions in which their products are manufactured and sold. Conformity to this standard does not, by itself, grant or enforce legal regulatory compliance.

9.7.1 Safety Standards

- TimeCard hardware is commonly evaluated against general electrical safety standards such as IEC 62368-1.
- Good engineering practices encourage the incorporation of over-voltage, over-current, and thermal shutdown thermal runaway protections.
- Required safety labels, high-voltage warnings, and chassis grounding instructions are typically made visible on the physical unit per market law.

9.7.2 Environmental Compliance

- Products sold into global markets generally conform to hazardous material directives such as RoHS, REACH, and WEEE.
- Hardware manufacturers are widely encouraged to utilize halogen-free PCB materials and design for extensive end-of-life recyclability.

9.7.3 End-of-Life Handling and Disposal

- It is recommended that vendors provide guidance for environmentally responsible hardware disposal.
- Best practices dictate that hazardous materials (e.g., lithium coin cells or backup batteries) are physically removed and sorted prior to e-waste recycling.
- A secure mechanism for the cryptographic erasure of firmware, private keys, and proprietary calibration data can significantly aid the secure decommissioning of units deployed in highly sensitive infrastructure.

9.7.4 Suggested Datasheet Inclusions

Comprehensive technical datasheets and integration manuals significantly reduce friction for end-users. Manufacturers are encouraged to document the following openly: - Complete environmental operating limits and a summary of qualification testing results. - Calculated MTBF and reliability datasets. - Proof of ESD and

EMC compliance certifications. - Required power, thermal dissipation (BTU/hr), and CFM airflow requirements. - Publicly available API/Register mappings to support unhindered multi-vendor interoperability.

10. Applications and Best Practices (Informative)

This chapter offers guidance on key deployment scenarios, application domains, and operational best practices for integrating and maintaining TimeCard systems. It helps system architects, integrators, and facility operators achieve optimal synchronization accuracy, stability, and reliability across diverse environments.

As an informative chapter, the guidelines presented herein are recommendations and do not constitute normative requirements for IEEE P3335 conformance.

10.1 Overview

TimeCards function as precision timing subsystems, providing highly stable, traceable, and interoperable time and frequency services to host platforms. Because the core architecture abstracts the complexities of hardware-level synchronization from the host CPU, TimeCards are versatile enough to be deployed across a wide range of industries.

Rigorous practices for physical installation, software configuration, and continuous monitoring help ensure that a TimeCard operates reliably at peak performance throughout its lifecycle.

10.2 Application Domains

The following subclauses detail the primary industries that leverage TimeCard architectures, noting the specific operational priorities and recommended configurations for each.

10.2.1 Data Centers and Cloud Infrastructure

- **Operational Purpose:** Coordinating distributed databases, globally synchronized transactions (e.g., Google Spanner/TrueTime equivalents), AI training cluster synchronization, and event logging across massive computing fleets.
- **Key Architectures:** PTP networks acting as the primary distribution method across the datacenter fabric, with TimeCards acting as Grandmaster clocks or high-precision Ordinary Clocks at the top-of-rack (ToR).
- **Recommended Best Practice:** Implement redundant TimeCards across multiple racks. Utilize PCIe Precision Time Measurement (PTM) to deterministically bridge the time from the network interface card directly to the host CPU domain, targeting sub-microsecond absolute accuracy.

10.2.2 Telecommunications and 5G/6G Networks

- **Operational Purpose:** Providing strict phase and frequency alignment for Open RAN (O-RAN) baseband units, fronthaul/backhaul cellular networks, and edge computing synchronization to prevent cellular interference.
- **Key Architectures:** Utilizing specialized telecom profiles such as ITU-T G.8275.1 (full timing support from the network) and G.8275.2 (partial timing support).
- **Recommended Best Practice:** Deploy a comprehensive ensemble of reference sources (e.g., GNSS augmented by network-delivered PTP). Operators are encouraged to actively monitor holdover Maximum Time Interval Error (MTIE) telemetry to anticipate network degradation during GNSS spoofing or jamming events.

10.2.3 Financial Systems and High-Frequency Trading (HFT)

- **Operational Purpose:** Enabling ultra-precise hardware timestamping for trading events to satisfy stringent regulatory compliance frameworks (e.g., MiFID II in Europe, SEC Rule 613 in the USA).
- **Key Architectures:** Direct 1PPS electrical signal distribution alongside high-frequency PTP broadcast networks, heavily reliant on hardware-timestamping at the exact point of ingress/egress.
- **Recommended Best Practice:** Maintain strict, mathematically provable traceability to Coordinated Universal Time (UTC) utilizing GNSS-disciplined master clocks. Calibration certificates should be maintained and refreshed annually to meet regulatory audit requirements.

10.2.4 Power Grid and Industrial Control Systems (ICS)

- **Operational Purpose:** Synchronizing distributed control systems, phasor measurement units (PMUs), SCADA networks, and high-voltage protection relays to monitor wide-area grid stability.
- **Key Architectures:** Utilizing the IEEE C37.238 PTP Power Profile over ruggedized deterministic Ethernet.
- **Recommended Best Practice:** Focus on environmental hardening. TimeCards deployed in substations typically require extended operating temperature ranges (-40°C to $+85^{\circ}\text{C}$) and robust electromagnetic interference (EMI) shielding to survive high-voltage switching transients.

10.2.5 Scientific Research and Metrology

- **Operational Purpose:** Distributing phase-coherent time signatures across particle accelerators, radio telescope arrays, and advanced metrology laboratories.
 - **Key Architectures:** Leveraging White Rabbit (WR) optical networks or specialized continuous-wave RF distribution (e.g., 10 MHz sine waves) to achieve sub-nanosecond precision.
 - **Recommended Best Practice:** Utilize high-stability local oscillators (such as Rubidium atomic clocks or high-end OCXOs). Operators are encouraged to continuously archive Allan Deviation (ADEV) and Time Deviation (TDEV) metrics to establish long-term baselines for experiment validation.
-

10.3 Deployment Best Practices

10.3.1 Hardware and Physical Installation

- **Thermal Management:** Precision oscillators are highly sensitive to thermal gradients. Installers are advised to verify that the host platform provides adequate, steady airflow across the TimeCard and to avoid placing the card adjacent to high-heat-dissipating components (e.g., flagship GPUs) un-baffled.
- **Cabling Quality:** For RF inputs and PPS outputs, operators are encouraged to utilize phase-stable, heavily shielded coaxial cables. Connectors (e.g., SMA) are best torqued to the manufacturer's exact specifications to maintain a consistent $50\ \Omega$ impedance boundary.
- **GNSS Antennas:** Route antenna cabling well away from high-noise switching power supplies or motorized equipment to minimize EMI ingress that degrades the signal-to-noise ratio.

10.3.2 Configuration of Synchronization Hierarchy

- **Source Prioritization:** Administrators are advised to configure a distinct reference hierarchy based on precision, stability, and availability (e.g., defining Local GNSS as Primary, Network PTP as Secondary, and adjacent 1PPS as Tertiary).
- **Ensemble Algorithms:** If the TimeCard supports multi-input ensemble modes, enabling this feature allows the card to mathematically weigh multiple high-quality references simultaneously, smoothing out transient anomalies from any single source.
- **Holdover Tuning:** Carefully tune holdover operational thresholds based on the characterized stability of the specific local oscillator model deployed, preventing the TimeCard from serving degraded time for too long after a reference loss.

10.4 Operational and Maintenance Best Practices

10.4.1 Monitoring, Telemetry, and Alarming

- **Continuous Observation:** Leverage out-of-band management interfaces (like IPMI, NC-SI, or dedicated I2C/SMBus bridges) to continuously poll synchronization health without burdening the host CPU.
- **Log Aggregation:** Stream frequency and phase offset telemetry into centralized time-series databases. This allows for the historical analysis of oscillator aging and network-induced jitter.
- **Threshold Alarms:** Configure proactive SNMP traps or Redfish events for critical state changes, such as reference loss (LOS), entry into holdover states, or sudden internal thermal spikes.

10.4.2 Firmware Lifecycle and Security

- **Cryptographic Verification:** Administrators are strongly advised to deploy only firmware packages that are cryptographically signed by the original hardware vendor.
- **Secure Staging:** Test all firmware updates on a staging TimeCard mirroring the production environment before executing fleet-wide deployments.
- **Access Control:** Restrict remote network access (e.g., REST/gRPC configuration ports) to isolated management VLANs, and routinely rotate authentication credentials.

10.4.3 Calibration and Traceability Maintenance

- **Routine Recalibration:** High-end OCXOs and atomic sources experience natural, slow frequency drift over years of operation (aging). Recalibrating the hardware against a primary standard (e.g., UTC(NIST)) minimizes this accumulated error.
- **Traceability Documentation:** Preserve and catalog the factory calibration certificates, as well as the logs of operational reference links, to simplify compliance audits in regulated industries.

10.5 Common Pitfalls and Mitigation Strategies

The following table outlines frequent deployment issues and recommended architectural or operational responses:

Issue / Symptom	Likely Root Cause	Recommended Mitigation Strategy
Constant 1PPS Misalignment	Reverse polarity settings or significant cable-length propagation delay.	Verify rising-edge vs. falling-edge configurations. Calibrate for cable propagation delay (~5 ns per meter).
Intermittent Reference Loss	GNSS antenna sky-view obstruction, or localized RF jamming/spoofing.	Implement a secondary network-delivered PTP reference as a fallback. Employ anti-spoofing GNSS receivers.
Excessive Drift in Holdover	Drastic thermal fluctuations affecting the oscillator during the holdover period.	Improve host chassis thermal regulation. Utilize a TimeCard model equipped with a higher-grade OCXO.
High Host-to-Card Latency	Relying on software interrupts and unoptimized OS networking stacks to fetch time.	Transition to in-band hardware timestamping mechanisms, such as PCIe PTM, to bypass OS scheduling jitter.

Issue / Symptom	Likely Root Cause	Recommended Mitigation Strategy
Management Interface Timeout	Host CPU exhaustion is preventing the OS from responding to in-band telemetry polls.	Shift telemetry polling to out-of-band (OOB) BMC channels, such as SMBus or NC-SI, that operate independently of the host OS.

10.6 Summary

By implementing the engineering, integration, and operational best practices detailed in this chapter, system architects mitigate the risks associated with distributing highly precise time across complex distributed systems.

These informative recommendations complement the normative hardware and logical requirements defined within the broader IEEE P3335 specification. When applied comprehensively, they enable the deployment of a robust, resilient, and highly traceable TimeCard infrastructure that performs reliably over decades of service.

Annex A: Metrics (Informative)

This annex provides definitions, mathematical measurement methodologies, and interpretation guidelines for the key timing and frequency metrics utilized throughout the IEEE P3335 TimeCard specification. Because this is an informative annex, the metrics and recommendations presented herein do not constitute strict normative requirements, but rather establish a common engineering vocabulary to facilitate consistent evaluation of stability, accuracy, and performance across different hardware implementations and test environments.

All metrics referenced in this annex heavily align with established industry metrology literature, specifically ITU-T Recommendations G.810 and G.8260, as well as IEEE Std 1139.

A.1 Overview

Metrology metrics quantify the operational performance of TimeCards across four primary domains: 1. **Time Stability:** The long-term coherence of the clock's phase. 2. **Frequency Stability:** The consistency of the clock's oscillator over specified averaging periods. 3. **Phase Noise:** The short-term, frequency-domain spectral purity of the oscillator. 4. **Holdover Behavior:** The predictable degradation of the timescale when external references are lost.

Utilizing a standardized mathematical framework allows operators to confidently compare devices from competing vendors and to validate interoperability across heterogeneous networks.

A.2 Core Timing and Stability Metrics

A.2.1 Allan Deviation (ADEV)

Definition: A statistical measure characterizing the fractional frequency stability of an oscillator or a timing system as a function of the averaging time (τ). Unlike standard variance, ADEV converges for common oscillator noise types (such as flicker frequency noise). **Formula:**

$$\sigma_y(\tau) = \sqrt{\frac{1}{2(N-1)} \sum_{i=1}^{N-1} (\bar{y}_{i+1} - \bar{y}_i)^2}$$

where $\bar{y}_i$ represents the continuous, successive fractional frequency averages measured over intervals of duration τ , and N represents the total number of frequency samples.

Purpose: ADEV is the primary metric for evaluating both short-term and long-term oscillator frequency stability. **Reference:** ITU-T G.810 Annex I; IEEE Std 1139.

Typical Baselines for Reference: | Oscillator Technology | Typical ADEV ($\tau = 1$ s) | Typical Application
 | |-----|-----|-----| | TCXO (Temperature-Compensated) | 1×10^{-9} | Cost-sensitive edge compute | | OCXO (Oven-Controlled) | 1×10^{-11} | Datacenter / Telecom boundary clocks | | Rubidium (Atomic) | 1×10^{-12} | Primary Reference Clocks (PRC) / Core | | CSAC (Chip-Scale Atomic) | 5×10^{-12} | Portable / SWaP-constrained field units |

A.2.2 Time Deviation (TDEV)

Definition: The time-domain equivalent of ADEV. It represents the measure of temporal stability related to the signal's phase variations. **Formula:**

$$\text{TDEV}(\tau) = \frac{\tau}{\sqrt{3}} \times \text{Modified ADEV}(\tau)$$

Purpose: TDEV specifically quantifies the variation in absolute time error over a given observation period. **Reference:** ITU-T G.810 Annex I. **Usage:** TDEV is highly useful for evaluating the precision of synchronization networks, particularly assessing packet delay variation (PDV) noise in PTP deployments.

A.2.3 Maximum Time Interval Error (MTIE)

Definition: The maximum peak-to-peak phase or time deviation observed between any two measurement points within an observation window of duration τ . **Formula:**

$$\text{MTIE}(\tau) = \max_{i,j} |x_j - x_i|, \quad \text{for all } (t_j - t_i) \leq \tau$$

where x is the time error constraint.

Purpose: MTIE acts as a worst-case boundary metric, measuring the absolute maximum wander or drift of the timescale. **Reference:** ITU-T G.8260 Appendix II.5. **Usage:** MTIE is universally utilized to define strict holdover performance boundaries (e.g., assessing if a clock drifts more than $1.5 \mu\text{s}$ over a 24-hour holdover period).

A.3 Frequency Domain and Jitter Metrics

A.3.1 Phase Noise ($\mathcal{L}(f)$)

Definition: The power spectral density of phase fluctuations of a continuous periodic signal, traditionally expressed in decibels relative to the carrier per hertz (dBc/Hz) at varying offset frequencies from the main carrier frequency. **Formula:**

$$\mathcal{L}(f) = 10 \log_{10} \left(\frac{S_\phi(f)}{2 \text{ rad}^2} \right)$$

where $S_\phi(f)$ is the one-sided spectral density of the phase deviations.

Purpose: Phase noise evaluates the short-term spectral purity of the local oscillator. **Reporting Profile:** Manufacturers typically plot phase noise across logarithmic decades: 1 Hz, 10 Hz, 100 Hz, 1 kHz, 10 kHz, and 100 kHz offsets. **Reference:** IEEE Std 1139.

A.3.2 Time Jitter

Definition: The short-term variance or instability of a time-domain signal (such as a 1PPS edge or a 10 MHz square wave) from its ideal, mathematically perfect periodic position. **Measurement Types:** - **Peak-to-Peak Jitter** (J_{pk-pk}): The absolute maximum time difference between the earliest and latest occurrence of the signal edge over the sample set. - **RMS Jitter** (J_{rms}): The root-mean-square average of the jitter samples, representing the common standard deviation of the timing error.

Measurement Best Practice: Jitter characterization is typically performed using calibrated Time Interval Counters (TICs) operating on a statistically significant dataset (e.g., 1×10^6 samples) with picosecond-level native resolution.

A.4 Accuracy, Precision, and Granularity

To prevent ambiguity in TimeCard documentation, developers are encouraged to consistently separate the definitions of accuracy, precision, and hardware granularity.

Metric	Contextual Definition	Example Target Profile
Accuracy	The degree of conformance or offset of the measured time directly to an absolute primary reference scale (e.g., UTC).	< 100 ns deviation from UTC(NIST) target
Precision	The internal repeatability or statistical variance of a measurement operation under completely identical conditions.	< 5 ns 1σ variance over 1,000 consecutive PTM transactions
Resolution	The absolute smallest physically distinguishable change the circuit can detect or measure.	10 ps inherent TDC (Time-to-Digital Converter) physical resolution
Granularity	The minimal mathematical discrete step available in the reported digital timestamps.	1 ns bit-level granularity within an IEEE 1588 packet payload

A.5 Holdover Metrics

When a TimeCard suffers a total loss of its inbound external reference signals (e.g., GNSS jamming or fiber cut), the active disciplining loop opens, and the unit transitions to **holdover**.

A.5.1 Holdover Time Error

Definition: The continuously accumulating deviation of absolute time driven by the free-running fractional frequency offset of the internal oscillator during holdover. **Mathematical Concept:**

$$\Delta x(t) = x_0 + \int_0^t \Delta y(\tau) d\tau + \frac{1}{2} D t^2 + \epsilon(t)$$

where x_0 is initial phase error, Δy is the normalized frequency offset, D is the linear frequency aging rate, and $\epsilon(t)$ represents internal random noise.

Practical Example: If a TimeCard enters holdover with an uncorrected frequency offset of 1.15×10^{-11} , the generated 1PPS signal will mechanically drift by approximately $1\mu\text{s}$ per day of holdover (≈ 86.4 ns per 1×10^{-12} offset).

A.5.2 Warm-Up and Recovery Behavior

- **Warm-Up Time:** The total duration required for an oscillator (specifically heated OCXOs or Rubidium cells) to reach thermal equilibrium and electrical stabilization upon a cold power cycle.
- **Recovery Time:** The architectural time required for the PLL to re-establish a solid lock and compress phase errors after successfully reacquiring a valid reference.
- **Note:** It is highly recommended that vendors empirically characterize and publish both parameters in standard hardware datasheets.

A.6 Ensemble and Correlation Metrics

In distributed environments, multiple TimeCards or reference inputs may be mathematically combined to form an **Ensemble Clock**.

Concept	Description
Ensemble Average Stability	Statistical improvement in composite ADEV proportional to $\sqrt{N}$, where N represents the number of completely independent, uncorrelated clocks operating in the ensemble.
Correlation Coefficient (ρ)	A statistical measurement quantifying the dependency or shared error vectors between clocks. A highly resilient ensemble relies on sources with low correlation ($\rho \approx 0$).
Weighting Function	The active algorithm determining the proportional contribution of each individual clock/reference step, dynamically shifting based on real-time MTIE performance or variance.

A.7 Environmental Influence Metrics

Because oscillators physically react to their environment, static laboratory metrics rarely reflect real-world holdover stability unless environmental parameters are factored in.

Parameter	Metric Definition	Impact
Temperature Coefficient	The variance in fractional frequency ($\Delta f/f$) per degree Celsius ($\Delta^\circ\text{C}$).	Directly dictates holdover drift if the chassis undergoes drastic thermal shifts (e.g., an AC failure).
Vibration Sensitivity (Γ)	The relative frequency change per g of mechanical acceleration.	Critical factor for systems deployed in heavily vibrating racks or industrial transit enclosures.

Parameter	Metric Definition	Impact
Aging Rate	The constant, predictable, unidirectional long-term frequency drift of the resonator over time (days/years).	Determines how frequently the TimeCard requires absolute external recalibration to prevent systemic clock bias.

A.8 Evaluating Hardware: Best Practices

When designing lab tests or automated qualification fixtures to evaluate the metrics detailed in this annex, engineers are encouraged to observe the following best practices: - **Traceability:** Utilize only highly calibrated measurement instruments (e.g., Phase Noise Analyzers, Counters) that possess verifiable traceability to primary national standards (NIST, PTB, NPL). - **Control Symmetries:** Rigorously design the physical measurement setup to minimize asymmetric cable propagation delays, thermal gradients, and RF impedance mismatches. - **Broad Spectrums:** Execute stability validation tests across broad averaging intervals ($\tau = 1$ s out to $> 10^5$ s) to capture both short-term jitter and diurnal temperature-driven wander. - **Contextual Logging:** Always log environmental metadata concurrently with performance data (e.g., ambient temperature, relative humidity, chassis airflow, and test equipment calibration schedules).

A.9 Summary

The mathematical metrics outlined in this annex establish a vital, unified vocabulary for characterizing the temporal performance of TimeCard architectures.

By standardizing exactly how stability, noise, drift, and holdover are mathematically calculated and reported, the IEEE P3335 ecosystem allows operators to rapidly compare discrete hardware configurations, validate regulatory compliance, and confidently integrate mixed-vendor timing infrastructure.

End of Annex A

Annex B: Test Procedures (Informative)

This annex provides standardized, repeatable engineering test procedures for evaluating the functional and performance characteristics of TimeCard implementations. These procedures are designed to ensure consistency, comparability, and mathematical traceability of measurement results across independent vendors, metrology laboratories, and live deployment environments.

As this annex is explicitly informative, the test methodologies proposed herein serve as best-practice engineering baselines for hardware validation rather than strict normative conformance requirements.

B.1 Overview

Comprehensive validation of a TimeCard architecture encompasses three primary testing domains: 1. **Functional Verification:** Validating logical compliance with the architectural interface, physical connectability, and management plane state-machine behaviors. 2. **Performance Validation:** Empirically measuring

TimeCard frequency stability, phase noise, holdover drift, and timestamping jitter against the vendor’s declared specifications. 3. **Environmental and Reliability Qualification:** Assessing physical performance consistency and survivability under externally applied, simulated stress conditions (e.g., thermal shock, vibration, or EMI).

To satisfy traceability requirements, it is strongly recommended that all tests be conducted utilizing equipment recently calibrated against recognized national metrology standards.

B.2 General Test Conditions

B.2.1 Environmental Setup

- Testing environments are typically maintained under nominal, thermally stable laboratory conditions unless a specific thermal extreme test is actively being executed. Baseline conditions generally sit at $23 \pm 3^\circ\text{C}$ with 30–70% non-condensing relative humidity.
- The Device Under Test (DUT) should be allowed to reach complete thermal equilibrium and internal oscillator operational stability (warm-up) prior to the commencement of any metrology recording.
- Physical cabling (e.g., coaxial cables for 1PPS or 10 MHz) and RF splitters utilized in the test fixture should be phase-matched and carefully impedance-matched (typically $50\ \Omega$) to prevent signal reflections that cause false jitter readings in the instrumentation.

B.2.2 Power and Initialization

- Evaluators should apply host bus power (e.g., via a PCIe riser) or auxiliary power within the hardware’s rated voltage tolerance.
- The initial power-up sequence provides a baseline to map the oscillator’s raw warm-up time from cold-start to achieving a steady-state frequency lock.

B.2.3 Measurement Equipment Calibration Limits

Instrumentation utilized in performance validation should possess a noise floor or error margin substantially lower than the DUT to ensure the DUT is being measured, rather than the noise of the instrumentation itself.

- **Reference Clock:** The primary laboratory reference clock driving the test equipment should possess an Allan Deviation (ADEV) at least one order of magnitude ($10\times$) superior to the target specification of the DUT.
- **Time Interval Counter (TIC):** TICs should possess a native single-shot resolution of $< 10\ \text{ps}$ for evaluating high-precision OCXOs or GNSS-disciplined TimeCards.
- **Phase Noise Analyzer:** The analyzer’s internal local oscillator should possess a phase noise floor at least 10 dB, preferably 15 dB, lower than the anticipated mask of the DUT across all relevant frequency offsets.

B.3 Functional Verification Tests

Functional tests validate the logical operations and state transitions of the TimeCard without necessarily grading sub-nanosecond physical precision.

B.3.1 Receive (Inbound) Interface Tests

1. **Reference Detection:** Sequentially apply valid external references (e.g., GNSS antenna input, PTP network feed, external 1PPS) and verify via the management telemetry that the TimeCard correctly recognizes the presence of the physical signal.
2. **Lock Acquisition:** Monitor the management plane to record the time taken for the TimeCard’s internal Phase-Locked Loop (PLL) or disciplining mechanism to transition from “Free-Run” to “Locked” status. Compare this to the vendor’s specified acquisition time limit.

3. **Failover Response:** While actively locked to a primary reference, abruptly disconnect the primary input. Verify that the hardware deterministically enters the “Holdover” state or smoothly transitions to a secondary reference without causing a systemic reboot or crashing the host driver.

B.3.2 Providing (Outbound) Interface Tests

1. **PPS Amplitude and Polarity:** Utilizing an oscilloscope bridged with a $50\,\Omega$ terminator, capture the 1PPS output edge. Verify the correct edge polarity (rising vs. falling), the slew rate (rise time), and the high/low voltage thresholds against standard CMOS/LVTTL specifications.
 2. **PCIe PTM Validation:** If supported, execute host-side software to trigger Precision Time Measurement (PTM) dialogs across the PCIe bus. Correlate the returned PTM hardware timestamps against the physical 1PPS output to verify systemic, in-band latency and offset behavior.
-

B.4 Performance Validation Tests

Performance tests quantify the TimeCard’s physical metrology capabilities.

B.4.1 Frequency Stability (ADEV/TDEV)

- **Procedure:** Connect the TimeCard’s primary continuous clock output (e.g., 10 MHz) to a Phase Noise Analyzer or high-resolution Frequency Counter. Record continuous phase/frequency samples locked against the laboratory primary reference.
- **Analysis:** Calculate and plot the Allan Deviation ($\sigma_y(\tau)$) and Time Deviation (TDEV) across logarithmic averaging intervals ($\tau = 1\text{ s}, 10\text{ s}, 100\text{ s}, 1,000\text{ s}, 10,000\text{ s}$).
- **Evaluation:** Compare the resulting stability curves against the manufacturer’s published datasheet limits.

B.4.2 Short-Term Phase Noise ($\mathcal{L}(f)$)

- **Procedure:** Utilize a cross-correlated Phase Noise Analyzer to plot the Single Sideband (SSB) phase noise of the oscillator.
- **Analysis:** Measure the power spectral density in dBc/Hz at standard decade offsets extending from 1 Hz to 1 MHz from the primary carrier frequency.
- **Evaluation:** Validate that the plotted noise curve sits below the vendor’s declared upper-bound phase noise mask.

B.4.3 Time Jitter

- **Procedure:** Capture a statistically significant sequence of 1PPS pulses (e.g., minimum 1×10^5 samples) utilizing a TIC triggered against the pristine laboratory 1PPS reference.
- **Analysis:** Calculate the Root-Mean-Square (RMS) jitter and the absolute peak-to-peak (J_{pk-pk}) timing variance of the signal edges.

B.4.4 Holdover Boundary (MTIE)

- **Procedure:** Allow the DUT to lock to a master reference and achieve thermal and mathematical equilibrium for at least 24 hours. Abruptly sever the reference connection to force the unit into holdover.
 - **Analysis:** Continuously record the wandering time error of the TimeCard’s 1PPS output against the laboratory master 1PPS for the specified holdover duration (e.g., 4, 12, or 24 hours).
 - **Evaluation:** Compute the Maximum Time Interval Error (MTIE) across the observation window and verify the drift profile remains beneath the target limit (e.g., $1.5\mu\text{s}/24\text{hr}$).
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B.5 Environmental and Stress Qualification

Environmental tests validate that the TimeCard’s hardware design robustly compensates for external physical interference.

Test Profile	Recommended Methodology	Verification Target
Thermal Cycling	Place the active, locked DUT inside an environmental chamber. Cycle the ambient temperature across the manufacturer’s maximum specified operating range (e.g., 0°C to 55°C) using a predefined ramp rate (°C/min).	Assess the maximum frequency excursion during temperature gradients. Verify that the oscillator’s internal temperature compensation mechanism functions correctly.
Power Interruption	Mechanically or electrically interrupt the host power plane for brief intervals (1 s to 10 s) if the TimeCard relies on localized holdover capacitors or batteries.	Verify that the hardware successfully bridges the interruption and maintains the internal unified timescale without experiencing a phase step or discontinuity upon power restoration.
Vibration / Shock	Subject the TimeCard to variable-frequency, multi-axis harmonic vibration (5 Hz to 500 Hz) using a specialized shaker table.	Record the immediate degradation in phase noise and frequency stability caused by g-force acceleration (Γ), verifying that the structural design adequately dampens the oscillator.

B.6 Reporting and Traceability

To maintain engineering transparency, comprehensive test reports for TimeCard validations should document:

- A detailed block diagram describing the physical test fixture, including cable lengths to account for fixed propagation delays.
- Expiration dates and calibration certificates for all primary laboratory measurement instrumentation used during the test.
- The precise firmware revision, hardware board step, and baseboard management controller (BMC) version installed on the DUT.
- Raw and plotted measurement data for all ADEV, TDEV, Phase Noise, and MTIE evaluations.
- Explicitly stated conditions during holdover tests (e.g., the thermal ramp rate applied to the chassis while the reference was disconnected).

With a consistent, traceable testing approach, integrators can confidently compare TimeCard implementations across different vendor ecosystems and ensure reliable behavior in production.

End of Annex B

Annex C: Bibliography (Informative)

This bibliography lists standards, engineering references, academic publications, and community resources that provide background for the design, testing, and implementation of TimeCard architectures.

These documents are non-normative. Their implementation is not required to claim conformance to IEEE P3335 unless a normative clause separately cites a document listed in Clause 2.

C.1 International Standards and Industry Profiles

- [B1] CENELEC EN 55032 / EN 55035, *Electromagnetic compatibility of multimedia equipment—Emission and Immunity Requirements*, 2017.
- [B2] ETSI EN 300 019 Series, *Environmental Engineering (EE); Environmental conditions and environmental tests for telecommunications equipment*.

- [B3] IEC 60068-2-6, *Environmental testing–Part 2-6: Tests–Test Fc: Vibration (sinusoidal)*, 2007.
- [B4] IEC 60068-2-27, *Environmental testing–Part 2-27: Tests–Test Ea and guidance: Shock*, 2008.
- [B5] IEC 61000-4-2, *Electromagnetic compatibility (EMC)–Part 4-2: Testing and measurement techniques–Electrostatic discharge immunity test*, 2009.
- [B6] IEC 62368-1, *Audio/video, information and communication technology equipment–Part 1: Safety requirements*, 2018.
- [B7] IEEE Std C37.238-2017, *IEEE Standard Profile for Use of IEEE 1588 Precision Time Protocol in Power System Applications*, 2017.
- [B8] ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories*, 2017.
- [B9] ITU-T Recommendation G.8275.1, *Precision time protocol telecom profile for phase/time synchronization with full timing support from the network*, 2016.
- [B10] ITU-T Recommendation G.8275.2, *Precision time protocol telecom profile for phase/time synchronization with partial timing support from the network*, 2017.
- [B11] ITU-T Recommendation G.8271, *Time and phase synchronization aspects of telecommunication networks*.
- [B12] Telcordia SR-332, *Reliability Prediction Procedure for Electronic Equipment*.

C.2 Academic and Technical Metrology References

- [B13] Allan, D. W., *Time and Frequency (Time-Domain) Characterization, Estimation, and Prediction of Precision Clocks and Oscillators*, IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control, 1987.
- [B14] Bregni, S., *Measurement of Maximum Time Interval Error for Telecommunications Clock Stability Characterization*, IEEE Transactions on Instrumentation and Measurement, Vol. 45, No. 5, 1996.
- [B15] Calhoun, J., *Clock Stability Characterization for Network Synchronization*, IEEE Communications Surveys & Tutorials, 2009.
- [B16] Lombardi, M. A., *NIST Technical Note 1952: Introduction to PTP and synchronization techniques*, National Institute of Standards and Technology, 2016.
- [B17] Riley, W. J., *NIST Special Publication 1065: Handbook of Frequency Stability Analysis*, National Institute of Standards and Technology, 2008.
- [B18] Weiss, M., *Time Transfer and Synchronization for Distributed Systems*, NIST Technical Report, 2013.

C.3 Supporting Open Source and Community Resources

- [B19] CERN / GSI, *White Rabbit Specification and Documentation*.
- [B20] Network Time Foundation, *NTP.org Reference Implementation source code and documentation*.
- [B21] Open Compute Project (OCP) Time Appliances Project (TAP), *TimeCard Hardware Schematics and Firmware Repository*. [<https://github.com/opencomputeproject/Time-Appliance-Project>]
- [B22] Open Compute Project (OCP) Time Appliances Project (TAP), *Interoperability Whitepaper and Multi-Vendor Testing Framework*.

C.4 Related Military/Defense Specifications

- [B23] DI-IPSC-81431A, *System/Subsystem Specification (SSS) Data Item Description*, 2000.
- [B24] DI-IPSC-81432A, *System/Subsystem Design Description (SSDD) Data Item Description*, 1999.
- [B25] DI-IPSC-81434A, *Interface Requirements Specification (IRS) Data Item Description*, 1999.
- [B26] DI-IPSC-81436A, *Interface Design Description (IDD) Data Item Description*, 1999.